

KATA Environment Architecture

Knowledge-Aware Technician Assignment

RL Agent (Transformer)

Policy

Embedding

action

obs (token_ids)

reward

KataEnv (Gymnasium Environment)

Observation
Builder

Value
Bucketing

State
Tokenizer

Reward
Composer

MCA
Warmup

MCA
Encoder

structured | tokens | token_ids

key: "SIM_TIME" val: "T_500_1K"

repair requests

action

SimPy Discrete-Event Simulation
GymTechDispatcher

repair queue + job orchestration

Repair
Queue

PreemptiveResource
(per tech)

assign

status

fit()

GymTechnician (x N)

Fatigue

Knowledge
Grid

Busy

Machines

Breakdown: Simple | Weibull

Machine
(simple)

ComplexMachine
(components)

request_repair()

repair()

Production Network

Product route: CNC -> Assembly -> Sink

Source

Router

knowledge
update
Feeder

Buffer

Sink

products

RepairRequest

machine_type, component_type
component_id, repair_time

RequestEncoder

Hash | Lookup | MCA

encode()

KnowledgeGrid

(ongoing library)
Gaussian propagation

KATAConfig

JSON | env vars | Python